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Developing regional soil micronutrient management strategies through ensemble learning based digital soil mapping

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ABSTRACT

Mapping of soil micronutrient variability is critical for improving agronomic biofortification. This study used 1778 surface soil samples collected from four agro-climatic regions of the Indo-Gangetic Plain of India to produce digital soil maps of available Zn, Cu, Fe, and Mn using 52 environmental covariates at a resolution of 150 m. The micronutrient prediction accuracy was compared for 14 machine learning approaches and their ensemble model. The hybrid ensemble model outperformed all 14 base learners and was subsequently used for producing micronutrient maps. All four micronutrients exhibited sufficient spatial variability. Both available Zn and Fe maps exhibited lower prediction uncertainties. Moreover, the inter-relationship between micronutrient concentration in soil and rice grain was explored to understand the Zn and Fe biofortification potential. The linear regression models revealed moderate agreement between soil available and grain micronutrient concentrations, with R^2 values of 0.52–0.63 for Zn and Fe, respectively. The developed models were used to predict grain Zn and Fe content from their respective soil concentrations, indicating the potential of the tested approach to identify specific pockets where rice varieties with biofortification potential can be planted. In the future, the digital soil mapping approach tested herein can help policymakers with regional decision-making, encouraging nutrient-based subsidy and investment opportunities and sustainable micronutrient recommendations toward micronutrient-enriched food. Further research is needed to develop a digital soil intelligence platform using micronutrient DSM products in resource-poor countries.

1. Introduction

Comprehensive knowledge of spatial variability of soil attributes helps improve agricultural management interventions to effectively manage soil nutrients (Minasny and McBratney, 2016). Widespread soil micronutrient deficiency reduces crop yields and adversely affects human and animal health (Saha et al., 2019). Indian soils are gradually becoming deficient in major cationic micronutrients, with ~ 49%, 33%, 5%, and 3% deficiency found in Zn, Cu, Fe, and Mn, respectively (Singh, 2008). This causes a reduction in plant growth and crop organ malfunction. Deficiency in soil micronutrients is also linked to a deficiency in humans (Yang et al., 2007). Globally approximately two billion people, constituting ~ 30% of the world's population, are affected due to micronutrient-deficient foods resulting in increased morbidity and mortality (FAO, 2013; Prom-u-thai et al., 2020). People in south Asian countries, including India, Sri Lanka, and Pakistan, have the lowest daily dietary intake of Zn (Wuehler et al., 2005). While Zn-deficient soils result in low Zn intake through foods and deficiency in humans (Zia et al., 2020), elevated Zn concentrations in food can also be harmful.

Agronomic biofortification is a promising and inexpensive strategy for improving micronutrient concentration in edible crop parts by application of soil and/or foliar fertilizers through boosting of mobility

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of soil micronutrients for better crop uptake (Szerement et al., 2021). Multi-country field trials have demonstrated the effectiveness of agronomic biofortification in food crops such as rice, wheat, and maize without compromising yield (Ram et al., 2016; Cakmak et al., 2017). Consequently, agronomic biofortification has proven to be a feasible and cost-effective strategy on a large scale.

The relationship between micronutrient bioavailability and the surrounding environment is not well established (Bowell and Ansah, 1993). Extensive deficiencies of micronutrients in crop production can be linked to intensive cropping and subsequent heavy mining of micronutrients from soils without replenishment (Moinuddin et al., 2017). Rice (*Oryza sativa* L.) is southeast Asia's most consumed edible food. Thus, agronomic biofortification can be a target for rice, primarily due to its inherent low concentration of micronutrients, varying nature of the re-translocation process from the vegetative parts to grains, and extensive losses during its processing stages (Ito and Lacerda, 2019). An elevated micronutrient pool is necessary for rice, especially before and/or during the grain-filling stage, for increasing grain micronutrients (Cakmak and Kutman, 2018). Therefore, assessing available micronutrient status in soil and correcting deficiencies is vital for increasing crop productivity. However, conventional spectrophotometric estimations of soil micronutrients are time-consuming, require skilled technicians, are costly, and need a specially designed laboratory set up with consumables (Wang et al., 2015). Accordingly, soil micronutrient testing facilities are not accessible to many farmers, especially in developing and resource-poor countries. Colorimetric soil sensors are inexpensive and available to farmers but lack the resolution need to effectively manage farms at a large scale. Moreover, site-specific nutrient management demands mapping of spatial variability for available soil micronutrients, which is not feasible solely with laboratory-based soil testing methods.

Conventional soil maps are static, and periodically reproducing them is cumbersome since their production is not well documented, apart from the unavailability of adequate spatial soil information and related *meta*-data (Lagacherie and McBratney, 2006). Nonetheless, the increasing availability of environmental covariate data such as digital elevation models, world climatic databases, land use, vegetation parameters, geomorphometric features, and climatic features (Hijmans et al., 2005), offer significant advantages in digital soil mapping (DSM) at various spatio-temporal scales (Pelegrino et al., 2022; Peng et al., 2019). Using representative samples measured in the ground, DSM exploits the associations between soil observations and environmental features. As summarized in a recent DSM review by Chen et al. (2022), in addition to geostatistical models, machine learning models or hybrid models (machine learning plus geostatistical models) are increasingly used for DSM studies.

So far, DSM interventions have been mainly confined to soil organic carbon (SOC) and few other soil parameters due to their low analytical cost and availability of legacy databases (Sreenivas et al., 2016; Dash et al., 2021; Arrouays et al., 2021). Nevertheless, some investigations have also focused on new covariates associated with soil attributes, especially those related to chemical features. For example, pH and soil texture can indicate the influence of topography and vegetation (Zhu et al., 2014). Consequently, there is a need to explore a diverse range of environmental covariates to be incorporated into future DSM activities. In developing countries, the DSM activities are severely hampered due to the lack of standardized digital soil databases, low sampling density, coarse data resolution, inadequate environmental covariates, few machine learning (ML) algorithms, limited independent validation, and non-availability of prediction uncertainty maps (Dash et al., 2021). Moreover, given DSM can map and study the interaction between soil attributes and environmental variables, a single ML algorithm might not provide sufficient information from the training data. Studies have shown that one algorithm may be able to map only some parts of the data set while the remaining part can be better explained using other algorithms (Tajik et al., 2020). Notably, the prediction for plant-available soil micronutrients using a wide assortment of environmental covariates and robust ML algorithms at a regional scale has remained largely unexplored. Besides, most DSM studies in countries like India have not presented uncertainty maps (Dash et al., 2021).

Ensemble learning implies a ML framework where multiple learners are trained to resolve the same problem (Zhou, 2009; Taghizadeh-Mehrjardi et al., 2019). Contrary to traditional ML approaches that aim to learn one assumption from the training dataset, a set of hypotheses is constructed and subsequently combined by the ensemble methods (Dietterich, 2002; Tajik et al., 2020). While a few recent studies have explored ensemble modeling for producing DSM products for soil class, SOC, and global soil mapping (Hengl et al., 2017; Song et al., 2020; Taghizadeh-Mehrjardi et al., 2019, 2020), the application of ensemble modeling for micronutrient mapping is still missing. Henceforth, a comprehensive study with diverse environmental covariates, such as climatic parameters, remote sensing information, organism-related covariates, and spectral indices, using the ensemble model with several ML models for plant-available soil micronutrients seems timely. This study explored a coordinated regional strategy to minimize the cost of extensive soil analysis for biofortification planning and judicious micronutrient fertilization. The objectives of the study were to: 1) compare the prediction accuracy of 14 ML approaches and their ensemble ability to optimally predict four available soil micronutrients, 2) identify the controlling factors of each soil micronutrient, 3) create high-resolution digital soil maps of four soil micronutrients (available Zn, Cu, Fe, and Mn – upper 30 cm) at the regional level with associated uncertainty at 90% prediction interval, and 4) develop a regional soil micronutrient management strategy using DSM generated maps. This is the first time a hybrid ensemble ML-based DSM technique has been developed and applied for spatial prediction and mapping available soil micronutrients. We hypothesized that DSM products using a hybrid ensemble model could help develop a regional soil micronutrient management strategy.

2. Materials and methods

2.1. Study region

The study was conducted in the Indo-Gangetic Plain (IGP), stretching across five Indian states (Shukla et al., 2020). The western part of IGP has a higher elevation than the eastern part. Generally, the IGP is characterized by a large floodplain deposited by a network of several rivers. The study region spans $>$ 45,000 km^2 and is divided into four distinct agro-climatic zones (Gangetic Alluvial Zone: GAZ, Coastal Saline Zone: CSZ, Vindhyachal Alluvial Zone: VAZ and Red and Laterite Zone: RLZ) over nine districts. The area extends from 24° 48′ 27″ N to 21° 41′ N latitude and 86°06′ 50′'E to 89° 25′ 38′'E longitude (Fig. 1). The elevation varies between −1 to 980 m from mean sea level. Average annual precipitation across the four agro-climatic zones ranges between 110 cm in the western RLZ to $\geq$ 161 cm in the CSZ. Mean annual maximum temperature is 33.8 °C to 35.6 °C. The CSZ features several islands with a mean elevation of 3.5 m above mean sea level and is surrounded by rivers connected to the Bay of Bengal. Consequently, the periodic intrusion of saline tidal water via rivers is a common feature of the CSZ (Sarkar et al., 2022). The study area covers almost half of the state of West Bengal, featuring mainly a rice-based cropping system (Mondal et al., 2020). West Bengal is one of India's leading rice-producing states and is also known as the richest reservoir of rice biodiversity and the country's rice bowl (Chatterjee et al., 2008). In West Bengal, rice is cultivated in both monsoon and winter seasons in the same field yearly, representing $>$ 60% of agricultural land holding. This rice-based mono-cropping system, adoption of similar soil-crop management practices, and non-addition of micronutrient fertilizers by small and marginal farmers result in the deficiency of available micronutrients in different parts of the rice growing zones. Other common agricultural and horticultural field crops are winter vegetables,

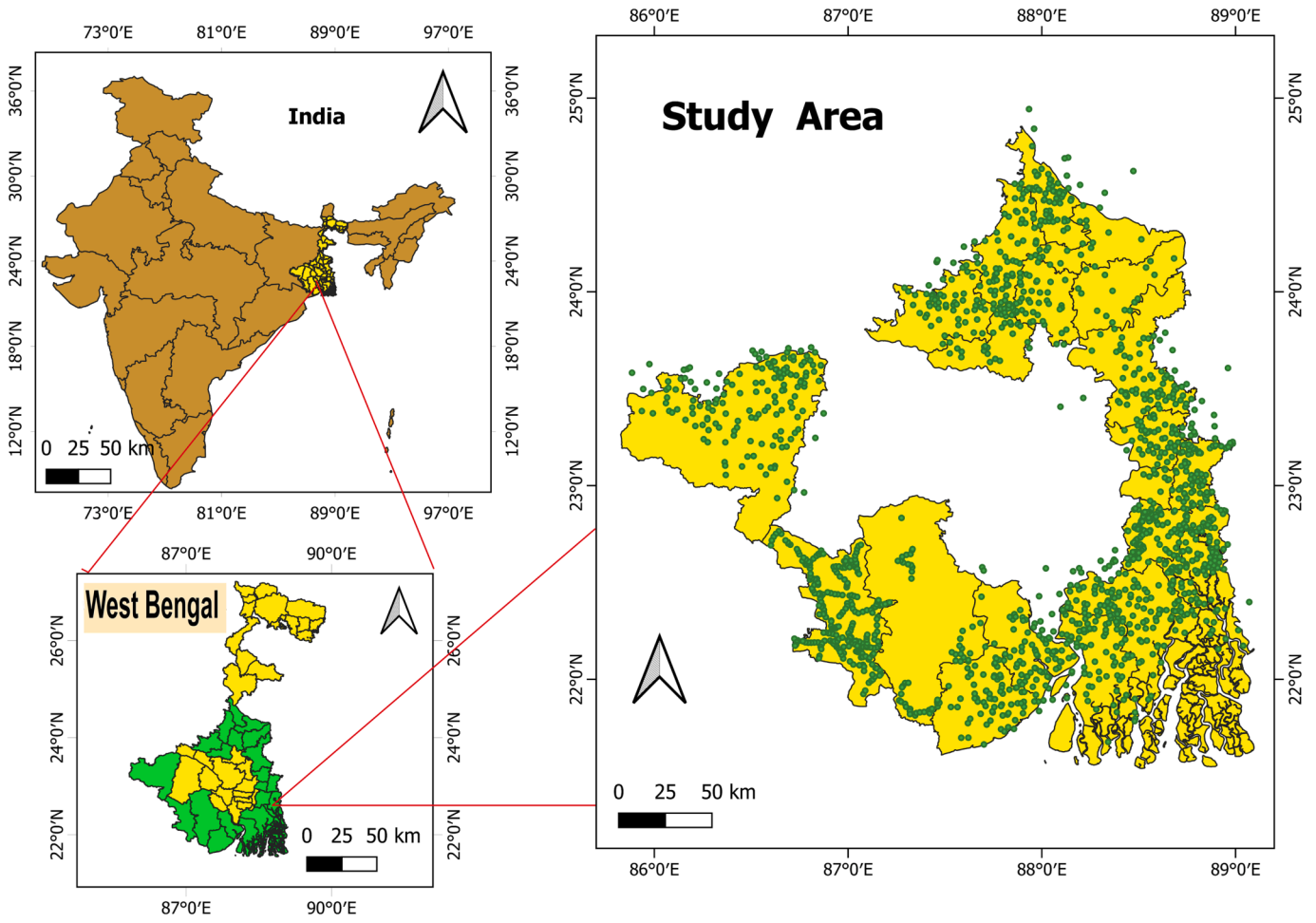


Fig. 1. Map of the study region in the Indo-Gangetic Plain.

pulses, potatoes, and groundnut (Dasgupta et al., 2022).

2.2. Soil and crop sampling and laboratory analysis

A total of 1778 surface (0–30 cm depth) soil samples were collected using an auger from the rice crop fields between 2017 and 2018 via a stratified random sampling approach (Fig. 1) considering the distribution of 68 soil series which adequately encompasses all variability among four agro-climatic zones featuring three distinct soil orders, and four soil parent materials. Approximately 25 soil samples were taken in each soil series, maintaining an average distance of 3 km. The number of sub-samples for making a representative composite sample was five to eight. Samples were collected during November-December after the harvesting of rice, from both rainfed and irrigated agricultural soils of small, medium, and large land holdings. Sampling points were well scattered throughout the region. Each sampling point was geolocated using a Garmin handheld GPS receiver (Garmin, Olathe, KS, USA) with an accuracy within ± 1 m. Soil samples were sealed in plastic Ziplock bags and transported to the laboratory for further analysis. Collected samples were distributed as follows: GAZ (n = 539), CSZ (n = 412), VAZ (n = 308), and RLZ (n = 519). In the entire soil library, 288, 637, and 583 samples represented different soil orders of Entisols, Alfisols, and Inceptisols, respectively. The samples were air-dried and subsequently ground and sieved using a 2-mm sieve. Soil pH was measured following the methodology of Soil Survey Staff (2004) using a 1:2 soil:water ratio via a digital pH meter (Systronics India, Model: 335). SOC was determined via chromic acid digestion (Walkley and Black, 1934). The concentrations of available Zn, Fe, Cu, and Mn in soil samples were determined by extracting soils (soil:solution = 1:2, v/v) with diethylenetriamine-penta-acetic-acid (DTPA) extractant (Lindsay and Norvell, 1978). The metal concentrations in the extracts were measured using an atomic absorption spectrophotometer (AAS) (Perkin Elmer model AAnalyst 400).

In the VAZ, several plant samples were collected apart from soil samples to establish a relationship between soil and rice crop grain micronutrient concentration. To evaluate the uptake of micronutrients (particularly Zn and Fe) by rice plants under natural conditions, while collecting samples, it was ensured that no samples were taken from fields receiving any micronutrient fertilizer to check the bioavailability pattern of micronutrients from only existing soil sources to rice grain and straw. A total of 48 sites were selected in a 70 km × 70 km area, covering all land holdings, from rainfed to irrigated, following the same sampling methodology used during soil sampling. Before harvesting, the composite crop samples from these 48 points were collected after one year of complete soil sample collection. Matured plant samples were collected from five to six nearby fields and composited for each site. Harvested plant samples were dried and separated into grain and straw. Both grain and straw were ground in a stainless-steel grinder and subsequently digested with a triacid mixture containing HNO_3, $HClO_4$, and H_2SO_4, (10:4:1, v/v). The total plant micronutrient (Zn and Fe) contents were analyzed via AAS as per Jackson (2005).

2.3. Environmental covariates for DSM

Various environmental attributes from multiple sources were used to represent the SCORPAN model (McBratney et al., 2003) for DSM of soil

micronutrient content. A total of 52 environmental covariates were used for the present study (Tables 1 and 2).

2.3.1. Terrain attributes

Terrain attributes were extracted from the Shuttle Radar Topographic Mission (SRTM) digital elevation model (DEM) at a spatial resolution of 30 m from Google Earth Engine (GEE). Subsequently, 20 DEM-based terrain-derived indexes were calculated using the SAGA-GIS version 6.3.0 (https://www.saga-gis.org). These attributes included elevation, aspect, analytical hill shading, convergence index, local downslope curvature, slope length and steepness factor (LS), topographic wetness index, topographic position index, valley depth, channel networks base level, channel networks distance slope, terrain ruggedness index, mid-slope, relative slope position, stream power index, SAGA wetness index system, multi-resolution ridge top flatness index, and multi-resolution valley bottom flatness index (Table 1).

2.3.2. Vegetation and soil attributes

Landsat 8 satellite imagery (30 m resolution) during November 2017 and 2018 (with $< 5\%$ cloud coverage) coinciding with the sampling time were extracted from the GEE. Subsequently, images were processed for their radiometric, geometric, and atmospheric corrections (Cooley et al., 2002; Mishra et al., 2014). A comprehensive range of soil, water, and vegetation indices (17 in number) was computed from the temporal median reflectance values corresponding to the Blue (B), Green (G), Red (R), near-infrared (NIR), shortwave infrared 1 (SWIR-1), and shortwave infrared 2 (SWIR 2) bands in different combinations (Table 3). Additionally, four spectral indices, normalized difference vegetation index (NDVI), soil adjusted vegetation index (SAVI), enhanced vegetation index (EVI) and green-normalized difference vegetation index (GNDVI) for describing vegetation growth conditions were computed. To account for the effect of parent materials and fertility in surface soil, nine other spectral indices such as bare soil index, carbonate index, clay index, salinity index, coloration index, redness index, hue index, grain size index, and brightness index were calculated (Zeraatpisheh et al., 2017). Additionally, four surface wetness indices such as land surface water index (LSWI), modified normalized difference water index (MNDWI), SWIR water stress index (SWSI), and water ratio index (WRI), were also included in the modelling framework since water content highly influences the chemistry and composition of certain soils and nutrient bioavailability (Morgan and Connolly, 2013). A common option for soil pH and SOC raster layer is GlobalSoilGrids at the 250 m $\times$ 250 m spatial resolution (Hengl et al., 2017). However, these layers were created from a global legacy soil database and may not be very accurate at a local scale, depending upon sampling density in the study area. Moreover, the layers may not highlight the current soil conditions as the legacy datasets have been collected at least a decade back. Therefore, in-situ measured soil pH and SOC content at sampled locations were interpolated into uniform spatial grids of 150 m $\times$ 150 m using the Kriging method. A spherical model with non-zero nuggets was used for fitting the variogram (supplementary material, Figs. SM-1 and 2).

2.3.3. Climatic variables

Climatic covariates were downloaded at 10 min ($\sim$340 km^2) spatial resolution from WorldClim version 2 data (https://worldclim.org). A total of 15 climatic covariates, known for their influence on soil properties, were included in the modeling and mapping (Table 2).

2.4. Selection of relevant features

In ML-based predictions, using redundant features often results in model overfitting. Therefore, feature engineering for selecting relevant features is a very important step for any ML algorithm. In the present study, the variable selection was executed using the Boruta feature ranking and selection algorithm through the R package "Boruta" (Kursa et al., 2010). Essentially, the Boruta algorithm is an extension of the

Table 1
Environmental covariates used for digital soil mapping of available soil micronutrients in West Bengal.

	Covariates	Description	Source	Software
Terrain	Elevation	Height above sea level (m)	SRTM DEM	SAGA GIS
	Aspect	The compass direction of the maximum rate of change	SRTM DEM	SAGA GIS
	Analytical hill shading	Tool for visualizing topography by simulating the earth surface illumination. It estimates surface reflectance from the sun at any altitude and any azimuth	SRTM DEM	SAGA GIS
	Slope	Average gradient above flow path	SRTM DEM	SAGA GIS
	LS factor	Slope Length and Steepness factor	SRTM DEM	SAGA GIS
	Local downslope curvature	This feature influences the rate of water flow through a slope which is a measure of soil and water movement and erosion		
	Valley depth	The vertical distance to a channel network base level	SRTM DEM	SAGA GIS
	Local curvature	The degree to which a curve deviates from a straight line	SRTM DEM	SAGA GIS
	Convergence index	Shows the structure of the relief as a set of convergent areas (channels) and divergent areas (ridges)	SRTM DEM	SAGA GIS
	Channel networks base level	The interpolated channel network base level elevations	SRTM DEM	SAGA GIS
	Channel networks distance	Providing information about the channel network by measuring relative height to the theoretical base of a channel network	SRTM DEM	SAGA GIS
	Topographic wetness index	Ln (specific catchment area/ slope angle)	SRTM DEM	SAGA GIS
	Topographic position index	Difference between a cell elevation value and the average elevation of the neighbourhood around that cell	SRTM DEM	SAGA GIS
	Terrain ruggedness index	It is a measure of local topographic relief. It calculates ruggedness as the sum of the elevation change within a user-specified neighbourhood	SRTM DEM	SAGA GIS
	Multi-resolution valley bottom flatness index (MRVBF)	Measure of flatness and lowness	SRTN DEM	SAGA GIS
	Multi-resolution ridge-top	Designed to identify valley bottom and high flat areas by	SRTM DEM	SAGA GIS

(continued on next page)

Table 1 (*continued*)

	Covariates	Description	Source	Software
	flatness index (MRRTF)	using slope classifications		
	Mid-slope	It covers the warmer zones of slopes	SRTM DEM	SAGA GIS
	relief position index	It is a quantitative measure of the degree of differentiation surface relief of the area	SRTM DEM	SAGA GIS
	Stream power index	Based on slope and specific catchment area	SRTM DEM	SAGA GIS
	SAGA wetness index System	For automated geoscientific analyses wetness index	SRTM DEM	SAGA GIS
	Total catchment area	An area from which surface runoff is carried away by a single drainage system	SRTM DEM	SAGA GIS
Climatic	Annual precipitation (mm)	The sum of daily rainfall in a year	Worldclim2	R studio
	Annual mean temperature	The approximate average of the maximum and minimum temperatures of the hottest and the coldest months of the year	Worldclim2	R studio
	Precipitation of driest month (mm)	The precipitation of the driest month	Worldclim2	R studio
	Precipitation of coldest month (mm)	The precipitation of the coldest month	Worldclim2	R studio
	Precipitation of wettest month (mm)	The precipitation of the wettest month	Worldclim2	R studio
	Precipitation of warmest month (mm)	The precipitation of the warmest month	Worldclim2	R studio
	Precipitation seasonality (%)	The Coefficient of Variation is the standard deviation of the monthly precipitation estimates expressed as a percentage of the mean of those estimates (e.g., the annual mean	Worldclim2	R studio
	Closed depression	A low, roughly concave topographic feature in a landscape.	Worldclim2	R studio
	Precipitation of driest quarter	The driest quarter of the year is determined (to the nearest month), and the total precipitation over this period is calculated	Worldclim2	R studio
	Precipitation of warmest quarter	The warmest quarter of the year is determined (to the nearest month), and the total precipitation over this period is calculated	Worldclim2	R studio
	Precipitation of coldest quarter	The coldest quarter of the year is determined (to the nearest month), and the total precipitation over this period is calculated	Worldclim2	R studio
	Mean temperature of coldest quarter	The coldest quarter of the year is determined (to the nearest month), and the mean temperature of this period is calculated	Worldclim2	R studio
	Mean temperature of wettest quarter	The wettest quarter of the year is determined (to the nearest month), and the mean temperature of this period is calculated	Worldclim2	R studio
	Mean temperature of driest quarter	The driest quarter of the year is determined (to the nearest month), and the mean temperature of this period is calculated	Worldclim2	R studio
	Mean temperature of warmest quarter	The warmest quarter of the year is determined (to the nearest month), and the mean temperature of this period is calculated	Worldclim2	R studio
Vegetation	Normalized difference vegetation index (NDVI)	It measures surface reflectance and also gives a quantitative estimation of vegetation status, growth and biomass	Landsat-8	GEE
	Soil adjusted vegetation index (SAVI)	A vegetation index that attempts to minimize soil brightness influences using a soil-brightness correction factor	Landsat-8	GEE
	Enhanced vegetation index (EVI)	Advanced vegetation index with improved sensitivity for high biomass regions used to quantify vegetation greenness through reduction in atmospheric influences and de-coupling of the canopy background signal	Landsat-8	GEE
	Green-normalized difference vegetation index (GNDVI)	Indicator for estimation of photo synthetic activity and it also well correlated with yield, plant growth, water and nitrogen uptake into the plant canopy	Landsat-8	GEE
Water	Land surface water Index	A measure of liquid water in vegetation canopies and its soil background by using the shortwave infrared (SWIR) and	Landsat-8	GEE

(*continued on next page*)

Table 1 (*continued*)

	Covariates	Description	Source	Software
		the NIR regions of the electromagnetic spectrum.		
	Modified normalized difference water index (MNDWI)	This index enhances open water features while suppressing noise from soil, vegetation and built-up land	Landsat-8	GEE
	SWIR water stress index	Significant absorption by leaf water occurs in the shortwave infrared wavelengths. Thus leaf water content is negatively related with plant reflectance. It is used as an indicator of canopy water stress	Landsat-8	GEE
	Water ration index	Used as a potential screening tool for separating water bodies from other land-cover features based on their spectral characters with green, red, NIR and MIR bands	Landsat-8	GEE
Soil	Grain size index (GSI)	It measures the close relationships between topsoil grain size and surface spectral reflectance. The reflectance patterns in the visible range bands can distinguish the topsoil with different grain size composition	Landsat-8	GEE
	Coloration index (CI)	Used as an indicator of soil colour	Landsat-8	GEE
	Redness index (RI)	Linked with iron oxide induced absorptions characteristics near 480 nm for goethite (typical yellow ± brown) and near 480 nm for goethite (typical yellow ± brown)	Landsat-8	GEE
	Hue index (HI)	Represents dominant wavelength	Landsat-8	GEE
	Bare soil index (BSI)	Used for the identification of bare soil areas and fallow lands from crop lands and others	Landsat-8	GEE
	Carbonate index (CI)	Spectral emissivity property of a carbonate rock composed of the two major carbonate minerals on the earth; calcite and dolomite	Landsat-8	GEE
	Clay index (ClI)	Hydroxyl groups of clay minerals typically absorb radiation at 2.2 μm and this feature is captured with SWIR clay index	Landsat-8	GEE
	Brightness index (GI)	Average soil reflectance	Landsat-8	GEE
	Salinity index (SI)	It is expressed to indicate the relationship between the actual soil salinity and the variation in vegetation, and for the surface reflectance of salt-affected land with sparse vegetation cover	Landsat-8	GEE
	Soil pH	Interpolated maps using Kriging method	In-situ	Q GIS
	Soil Organic Carbon	Interpolated maps using Kriging method	In-situ	Q GIS

random forest (RF) method and utilizes the importance measure generated by the original algorithm. It also compares the importance of original attributes with their randomized copies (Kursa et al., 2010). Recently, Boruta has gained popularity as a novel feature selection method in streamflow prediction (Ahmed et al., 2021), modeling forest biodiversity (Leutner et al., 2012), soil moisture forecasting (Prasad et al., 2019), and crop prediction (Suruliandi et al., 2021). The importance of all covariates was measured based on mean absolute error (MAE). Finally, all extracted covariates (30 m × 30 m) were rescaled and resampled to the cell size of 150 m × 150 m using bilinear interpolation in R 3.5.2 (R Core Development Team, 2020) via the caret package (Kuhn, 2022). Fig. 2 represents maps of four randomly selected terrain, climatic, remote sensing, and soil variables.

2.5. Machine learning based prediction of soil micronutrients

Over the last decade, ensemble ML method has gained popularity in almost all fields, including quantifying the soil spatial relationships. In the ensemble learning method, several algorithms were combined in a single model to maximize the prediction accuracy, which showed a higher prediction power than the individual models (Rojas et al., 2008; Malone et al., 2014). Subsequently, individual predictions from each base learner and their weighted average represent the final ensemble model. In this study, all feature selection and modeling were performed using the "caret" and "caretEnsemble" packages in R 3.5.2 (R Development Core Team, 2020). The box and density plots of the original dataset exhibited that the data had a wide range with some extreme outliers and were non-normally distributed for all the micronutrients. Consequently, the Tukey Krammer approach (Tukey, 1977) was implemented using the "transformTukey()" function from the R package rcompanion (Mangiafico, 2016), which iteratively uses the Shapiro-Wilk test for finding suitable λ value to convert the non-normally distributed data to normal. The optimal values of λ were 0.3 (Fe), 0.42 (Mn), and 0.6 (Zn and Cu). Notably, the transformed data appeared more normal than its original form and was used for model building.

This study used a model blending framework for spatial prediction of soil micronutrient content. At first, 14 base learner algorithms (Table 3), representing a broad and diverse set of ML models were used for predicting and mapping the spatial variability of four soil micronutrients (available Zn, Cu, Fe, and Mn) using the transformed data. The base learners included: least absolute shrinkage and selection operator (LASSO), multivariate adaptive regression splines (Earth), recursive partitioning and regression trees (rpart), generalized linear models (glm), support vector machines with radial basis function kernel (svmRadial), support vector machine (SVM) with linear kernel

Table 2
Soil and vegetation indices derived from the Landsat 8 spectral bands: Blue (B), Green (G), Red (R), near infrared (NIR), shortwave infrared 1 (SWIR 1), and shortwave infrared 2 (SWIR 2). DEM and SAGA GIS indicate digital elevation model and system for automated geoscientific analyses (SAGA GIS), respectively.

Index Name	Target Propriety	Wavelength and formula	Reference
Band 2	Soil	B (450–515 nm)	United States Geological Survey (2014)
Band 3	Vegetation	G (525–600 nm)	United States Geological Survey (2014)
Band 4	Vegetation with slope	R (630–680 nm)	United States Geological Survey (2014)
Band 5	Biomass content	NIR (845–885 nm)	United States Geological Survey (2014)
Band 6	Soil moisture and vegetation	SWIR 1 (1560–1660 nm)	United States Geological Survey (2014)
Band 7	Soil moisture and vegetation	SWIR 2 (2100–2300 nm)	United States Geological Survey (2014)
Normalized difference vegetation index (NDVI)	Health of vegetation	(NIR-R)/(NIR + R)	Forkuor et al. (2017)
Soil adjusted vegetation index	Soil moisture and vegetation	(NIR-R) *0.5/ (NIR + R + 0.5)	Huete (1988)
Enhanced vegetation index	Vegetation greenness	NIR-R/(NIR + 6*R-7.5*B + 1)	Huete et al. (2002)
Green-normalized difference vegetation index (GNDVI)	Crop photo synthetic activity	(NIR − G)/(NIR + G)	Gitelson et al. (1996)
Land surface water index	Liquid water in vegetation and its soil background	(NIR − SWIR)/ (NIR + SWIR)	Xiao et al. (2004)
Water ration Index	Separation for water bodies and non-water	(G + R)/(NIR + SWIR-1)	Shen and Li (2010)
SWIR water stress index	Canopy water stress		Fensholt and Proud (2003)
Modified normalized difference water index (MNDWI)	Water bodies	(G–SWIR1)/(G + NIR)	Xu(2006)
Grain size index (GSI)	Texture (grain size composition)	(R-B)/(R + B + G)	Xiao et al. (2006)
Coloration index (CI)	Soil colour	(R-G)/(R + G)	Srisomkiew, et al. (2022)
Redness index (RI)	Hematite content	R^2 /(B*G^3)	Mathieu et al. (1998)
Hue index (HI)	Primary colours	(2*R-G-B)/(G-B)	Mandal (2016)
Bare soil index	To discriminate bare soils	(SWIR 2 + R)-(NIR + B)/(SWIR 2 + R)+(NIR + B)	Diek et al. (2017)
Carbonate index	Dominant carbonate minerals on the earth; calcite and dolomite.	R/G	Abdel-Kader (2011)
Clay index		SWIR-1/SWIR-2	Gasmi et al. (2022)
Brightness index (BI)	Average reflectance magnitude	((R^2 + G^2 + B^2)/ 3)$^{0.5}$	Forkuor et al. (2017)
Salinity index	Salt-affected surface land	$(B)^{0.5}$ * R	Khan et al. (2001)

Table 3
Selected machine learning models and their hyperparameters used to model micronutrients in West Bengal soils.

Models	Definition	Hyperparameters	References
LASSO	The least absolute shrinkage and selection operator	lambda	Tibshirani (1996)
MARS	The multivariate adaptive regression splines	nprune, degree	Friedman et al. (2001)
rpart	Recursive Partitioning and Regression Trees	complexity parameter	Therneau and Atkinson (1997)
Rpart2	Similar to rpart and based on the same recursive partitioning, but uses different methods for optimization.	max tree depth	
svmRadial	Support vector machines with radial basis function kernel	Kernels, degree	Ding et al. (2021)
svmLinear	Support vector machines with linear kernel	cost	Ben-Hur et al. (2008)
Cubist	Cubist committees	neighbors	Quinlan (1992)
RF	Random Forest	mtry, ntree	Breiman (2001)
QRF	Quantile Regression Forests	mtry	Meinshausen and Ridgeway (2006)
XGBoost	Extreme gradient boosting e.g., booster,	Max depth, subsample, eta	Chen et al. (2015)
extraTrees	Extremely randomized trees	K, M	Geurts et al. (2006)
XGB	Gblinear booster of XGBoost family	Max depth, subsample, eta	Chen et al. (2015)
glmStepAIC	Generalized linear model with stepwise feature selection	–	
leapSeq	Linear regression with stepwise selection	maximum number of predictors	
Ensemble learning	Weighted average of multiple base models to produce one optimal predictive model	weights	Dietterich (2002); Polikar (2012)

(svmLinear), cubist, random forest (RF), quantile regression forests (QRF), extra-trees regressor (extraTrees), linear regression with stepwise selection (leapSeq) extreme gradient boosting (XGBoost), gblinear of the XGBoost family (xgbLinear), and generalized linear model with stepwise feature selection (glmStepAIC).

The permutation feature importance method was first used by Breiman (2001) to select the important variables in the RF algorithm and later was improved by Fisher et al. (2019) for use in interoperating ML methods. This method can be used in any ML algorithm and takes every predictor and permutes that predictor, observing how the prediction quality of that ML algorithm changes compared to the original ML prediction. Root mean square error (RMSE) is generally considered a performance metric with no permutation of variables. The more decrease in RMSE, the more significant that variable. In this study, a permutation feature importance using the VIP package in R 3.5.2 was applied to rank covariates based on their significance in the ensemble model of all four micronutrients. The learner's error ($error^{orig}$) was calculated based on Eq. (1):

$$error^{orig} = measured\ soil\ property - f\left(covariates^{orig}\right) \quad (1)$$

For every covariate, randomly permuted covariates ($covariates^{perm}$) were introduced to reduce their correlation with soil properties. Base learner's error (f) was calculated using Eq. (2):

$$error^{perm} = measured\ soil\ property - f(covariates^{perm}) \quad (2)$$

Moreover, permutation feature importance (PFI) was calculated

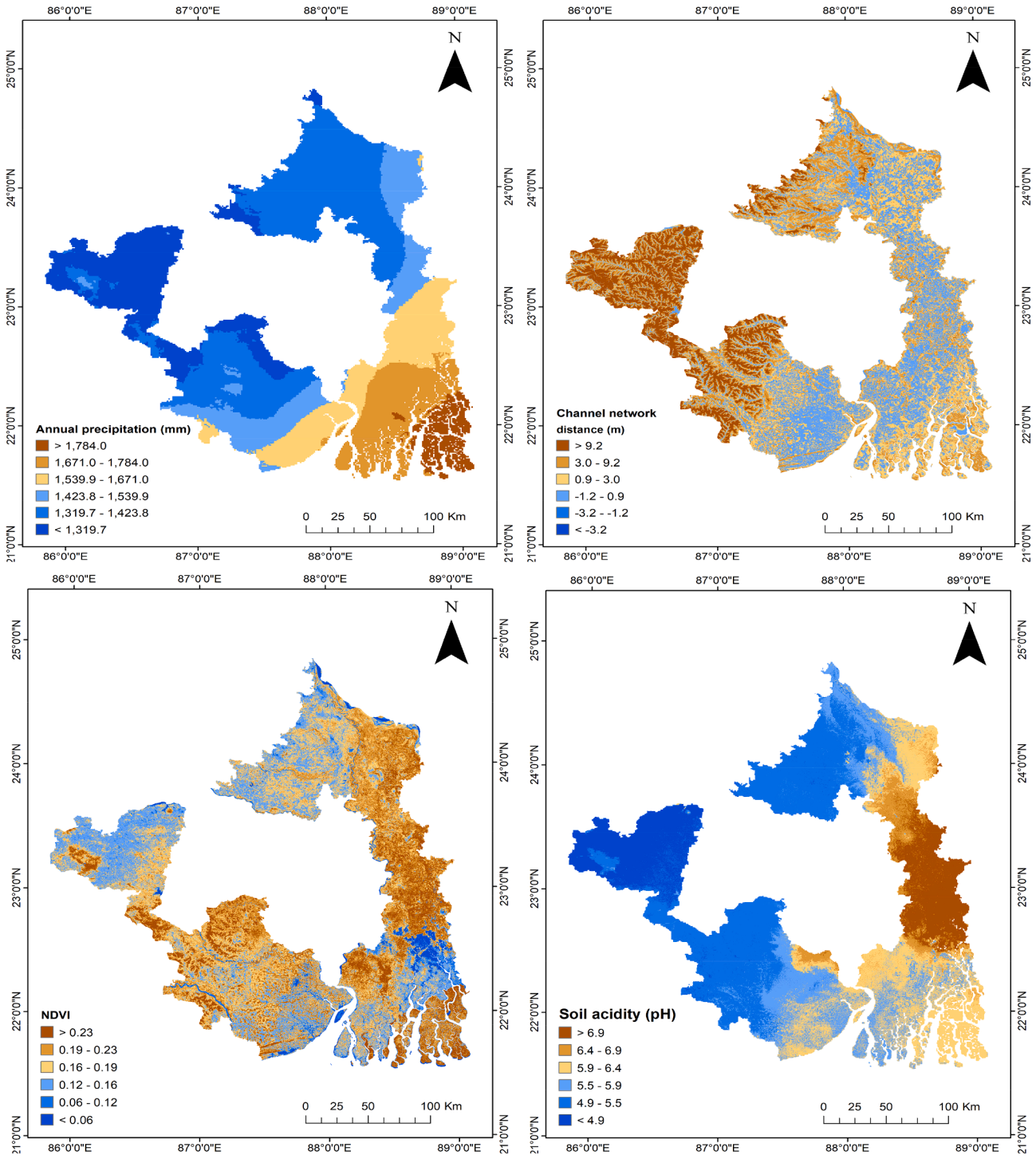


Fig. 2. Maps of four randomly selected climatic (annual precipitation), channel network distance (elevation), remote sensing (normalized difference vegetation index, NDVI), and soil variables (soil acidity) used for producing maps of micronutrients in the Indo-Gangetic Plain.

based on Eq. (3):

$$PFI = error^{perm}/error^{orig} \quad (3)$$

Eqs. (1) to (3) were repeated, and the covariates were sorted based on their influence in descending order. To calculate the contribution of each covariate in the final ensemble algorithm, the weight of each base learner was multiplied with descending order covariates.

Initially, the whole dataset was partitioned into a training dataset (70%) and a test dataset (30%), using a stratified random split. Next, the 14 base learners were trained five times using the training dataset via ten-fold cross-validation. For tuning the hyperparameters of each model, a tuneLength of 10 was set, which automatically does a grid search for suitable hyperparameters and selects the best model. Cross-validated outputs from 14 base learners were used for fitting a generalized linear model. Finally, the accuracy of the base learners and ensemble model was checked using the test dataset (30%). Notably, ensemble prediction was calculated using the weighted average based on PFI of Boruta-selected covariates.

2.6. Assessment of model performances and uncertainty

After prediction of the micronutrient content using ML models, the predicted values were back-transformed to the original scale. Prediction accuracy was then estimated using mean absolute percentage error (MAPE) (Eq. (4), RMSE (Eq. (5), and Lin's concordance correlation coefficient (ρc) (Eq. (6) (Lawrence and Lin, 1989).

$$\text{MAPE} = (1/n)*\Sigma(|y_i - \hat{y}|/|y_i|)*100 \quad (4)$$

$$\text{RMSE} = \sqrt[2]{\frac{1}{N}\sum_i^N \sum (y_i - \hat{y})^2} \quad (5)$$

$$\rho_c = 2Cov(X, Y)/\ Var(X) + Var(Y) + [E(X) - E(Y)]^2 \quad (6)$$

Above, n, y_i, and $\hat{y}$ indicate the number of observations, actual values, and estimated values, respectively. Moreover, X and Y indicate two random variables. E(X) and E(Y) denote the expected values of X and Y, respectively.

For this experiment, bootstrapping (Liddicoat et al., 2015) was used to quantify the uncertainty of soil micronutrient prediction. The full dataset was randomly split between calibration (70%) and validation (30%) sets. The bootstrapping analysis was performed only on the training set, while the 30% validation set was used for blind testing. The predicted maps with back-transformed original values were then used to calculate the mean and standard deviation of the prediction of each pixel. In this study, the uncertainty maps were generated only from the bootstrapping iterations (calibration set) from the ensemble model for each soil micronutrient and reflected a 90% confidence interval. However, it is noteworthy that the variance from the bootstrapping accounts for part of the uncertainty, while the remaining part is from the residual variance on the validation set. The sum of two variances is then used to determine the total uncertainty (Malone et al., 2016). All digital maps and uncertainty maps were generated using ArcMap version 10.2 (ESRI Inc., Redlands, CA, USA) via spatial analyst and geostatistical analysis extensions. Finally, the predicted map outputs were compared with rice crop grain concentrations collected from the same pixel location to validate the final model results.

3. Results and discussion

3.1. Descriptive statistics for soil micronutrients

Descriptive statistics for four analyzed soil micronutrients, using original vales, along with soil pH and SOC (%), are presented in Table 4. All micronutrients exhibited wide variability, indicating variation in parent material, geology, vegetation, local management practices, soil biota, and topography throughout the study area (Bhunia et al., 2018). In addition, the soil pH of the study area also varied widely (3.10–8.90) from strongly acidic to highly alkaline, suggesting the influence of soil reaction on spatial variability of micronutrients (Brady and Weil, 2002). Moreover, the SOC (%) of the sampled soils also showed wide variability (0.11–2.37%). Notably, the RLZ and VAZ exhibited mean SOC values of 0.61% and 0.63%, respectively, which resulted from continuous cropping without applying adequate organic matter in these soils over the years (Patra et al., 2014). In contrast, the CSZ and GAZ exhibited moderate to high SOC, with a mean of 0.96% and 0.82%, respectively. As expected, the coefficient of variation (CV) of the SOC for the four agro-climatic zones varied widely from 28 to 65%. The mean availability of the analyzed micronutrients followed the order: Fe > Mn > Cu > Zn. Given plant-available Zn is highly influenced by many factors, including climatic and topographic variability (Shuman, 1986), available Zn content for the moist sub-humid CSZ ranged from 0.05 to 8.94 mg kg^{-1} with a mean of 1.32 mg kg^{-1}, whereas in dry subhumid VAZ and hyperthermic RLZ; the mean values were comparatively higher (1.65 and 1.58 mg kg^{-1}, respectively). In the study area, relatively higher values of soil Cu were found in the GAZ and CSZ with a mean of 4.81 and 4.32 mg kg^{-1}, respectively, possibly due to comparatively higher SOC and soil pH (Laurent et al., 2020). Overall, the CV values of all four analyzed micronutrients indicated very high variability (>30%) as per the criteria proposed by Gomes and Garcia (2002) (Table 4). Similarly, Shukla et al. (2020) revealed the wide variability of soil available Fe, Mn, Cu, and Zn in other parts of India. The highest and lowest CVs obtained for Fe were 62.95% and 46.92% for the GAZ and CSZ, respectively. Conversely, for Mn, VAZ (82.41%) and RLZ (68.24%) exhibited maximum and minimum variability, respectively.

3.2. Prediction model performance

In this study, 20 covariates were extracted from 52 using Boruta and were subsequently used in the micronutrient prediction models (Table 5). Out of 14 base learners, the lowest validation RMSE was found in tree-based algorithms such as extraTrees, QRF, and RF (Fig. 3). The better performance of the tree-based algorithms can be attributed to the reduction of overfitting in decision trees that offers robust prediction, even for very high dimensional datasets (Biau, 2012). In comparison, the ensemble model exhibited the lowest validation RMSE values, as

Table 4
Descriptive statistics of measured soil chemical properties for 1778 collected soil samples from West Bengal.

Statistics	pH	SOC (%)	Zn (mg kg^{-1})	Cu (mg kg^{-1})	Fe (mg kg^{-1})	Mn (mg kg^{-1})	pH	SOC (%)	Zn (mg kg^{-1})	Cu (mg kg^{-1})	Fe (mg kg^{-1})	Mn (mg kg^{-1})
RLZ(N = 519)							**VAZ (N = 308)**					
Minimum	3.10	0.11	0.23	0.16	15.00	1.77	3.65	0.19	0.19	0.12	12.90	3.84
Maximum	7.50	1.63	10.84	15.19	454.00	168.28	7.48	2.37	8.05	8.17	330.60	191.20
Mean	5.04	0.61	1.58	2.99	122.00	39.10	5.34	0.63	1.65	2.87	96.74	33.09
Median	5.06	0.57	1.256	2.623	110.26	3.287	5.1	0.495	1.43	3.13	84.8	31.4
Std. Deviation	0.75	0.40	1.48	1.95	73.20	26.68	0.78	0.30	1.15	1.83	56.12	27.27
Variance	0.56	0.16	2.18	3.82	5358.82	711.60	0.61	0.09	1.33	3.33	3149.07	743.60
Skewness	0.09	11.39	4.47	1.47	1.42	1.39	0.57	1.77	2.62	0.54	1.33	2.94
Kurtosis	−0.02	198.04	27.66	4.14	2.74	2.41	0.08	5.06	10.69	−0.49	1.96	12.67
CV (%)	14.88	65.57	93.67	65.22	60.00	68.24	14.61	47.62	69.70	63.76	58.01	82.41
GAZ (N = 539)							**CSZ (N = 412)**					
Minimum	4.33	0.17	0.29	0.23	7.26	0.88	4.10	0.26	0.15	0.15	12.32	1.62
Maximum	8.90	1.85	24.70	40.75	373.63	128.30	8.90	2.19	8.94	9.84	402.40	209.40
Mean	7.19	0.82	1.55	4.81	92.30	25.46	6.15	0.96	1.32	4.32	154.99	36.70
Median	7.4	0.84	0.995	3.85	78.92	19.89	6.10	0.93	1.10	4.52	155.84	30.88
Std. Deviation	0.86	0.23	4.26	3.90	58.10	19.60	0.88	0.32	0.91	2.01	72.72	26.30
Variance	0.74	0.05	18.18	15.23	3375.97	384.34	0.77	0.10	0.83	4.05	5287.49	691.54
Skewness	−1.04	−0.01	10.59	2.88	1.50	1.69	0.44	0.63	3.02	−0.02	0.12	2.21
Kurtosis	0.74	−0.10	116.86	18.47	2.87	3.89	0.01	0.81	15.66	−0.52	−0.74	8.00
CV (%)	11.96	28.05	274.84	81.08	62.95	76.98	14.31	33.33	68.94	46.53	46.92	71.66

Table 5
The results of Boruta selected 20 covariates used for each soil micronutrients.

Sl No.	Zn (mg kg^{-1})	Cu (mg kg^{-1})	Fe (mg kg^{-1})	Mn (mg kg^{-1})
1	Precipitation Seasonality	Precipitation Seasonality	pH	Enhanced Vegetation Index (EVI)
2	Mean Temperature of Warmest Quarter	Annual Precipitation	Annual Precipitation	NDVI
3	Mean Temperature of Wettest Quarter	MRRTF	SOC	Annual Mean Temperature
4	Bare Soil Index	pH	Precipitation of Driest Quarter	pH
5	Precipitation of Wettest Quarter	Precipitation of Warmest Quarter	Mean Temperature of Wettest Quarter	Mean Temperature of Driest Quarter
6	Annual Precipitation	Mean Temperature of Driest Quarter	Precipitation Seasonality	Precipitation of Driest Quarter
7	Precipitation of Warmest Quarter	Mean Temperature of Warmest Quarter	Mean Temperature of Warmest Quarter	Annual Precipitation
8	Mean Temperature of Coldest Quarter	Channel Network Base Level	MRRTF	Channel Network Base Level
9	Mean Temperature of Driest Quarter	Precipitation of Driest Month	Precipitation of Coldest Quarter	Hue Index
10	Precipitation of Driest Quarter	MRVBF	Annual Mean Temperature	SOC
11	Precipitation of Coldest Quarter	Annual Mean Temperature	Mean Temperature of Driest Quarter	Coloration Index
12	Precipitation of Driest Month	SOC	Precipitation of Wettest Month	Modified NDWI
13	Precipitation of Wettest Month	Elevation	Enhanced Vegetation Index (EVI)	Grain Size Index
14	Channel Network Base Level	Enhanced Vegetation Index (EVI)	NDVI	Carbonate Index
15	Coloration Index	NDVI	Analytical Hillshading	Local Downslope Curvature
16	Annual Mean Temperature	Slope	MRVBF	Bare Soil Index
17	Carbonate Index	Clay Index	Hue Index	Channel Network Distance
18	Channel Network Distance	LS factor	Slope	Clay Index
19	Clay Index	Saturation Index	Land Surface Water Index	LS factor
20	Aspect	Salinity Index	Bare Soil Index	Aspect

indicated by the red dashed line (Fig. 3). Subsequently, the ensemble model was used for predicting micronutrient concentrations and yielded validation RMSE values of 0.44, 1.22, 22.34, and 6.42 mg kg^{-1} for Zn, Cu, Fe, and Mn, respectively, using back-transformed values (Fig. 4). The relatively better performance of the ensemble model can be ascribed to three reasons. First, when the training data is not enough for an algorithm to find the inner relationship between the dataset, the ensemble model performs better than the individual algorithm. Secondly, a particular algorithm might have some assumptions that cannot be satisfied. Thirdly, the ensemble model can offer better results in the hypothesis space, where an algorithm searches for a solution that may not contain a valid target (Tajik et al., 2020).

Admittedly, for obtaining a quick solution for a predictive model, it is not necessary to fit a large ensemble model. However, different models offer different perspective (and interpretation) of the underlying unknown truth. In addition, by using many approaches, the investigator can compare their performance and have confidence. Summarily, an ensemble model can offer a holistic view of the truth and improve prediction performance as a side effect. Finally, with the advancement of computing power, it is much easier to fit several models quickly, simultaneously, and efficiently by using parallel programming, graphics processing unit (GPU) programming, etc. In the future, it will be much easier to compute many models and combine them efficiently.

Apart from soil pH, the permutation feature importance plots of the ensemble model revealed the dominance of climatic variables like precipitation seasonality, annual precipitation, precipitation of driest and wettest quarter, mean temperature of driest and wettest quarter, etc., for all four micronutrients (Fig. 5). Collectively, these indices represent either variable soil water stress or water availability situations. Similarly, these indices represented the alternate wetting and drying condition, a distinct climatic characteristic of Southeast Asia, causing short-term changes in soil pH and redox conditions, significantly altering the form and bioavailability of micronutrients (Bowell and Ansah, 1993). Notably, adverse soil moisture and temperature influences can reduce soil micronutrient bioavailability and concomitant plant uptake (Alloway, 2009). Climatic variables were followed by either remote sensing indices, topographic, or SOC for different micronutrients.

Under a tropical, intense leaching environment, topography-driven erosional processes accelerate soil nutrient transport through rainwater and sediment across the slope (Hounkpatin et al., 2022). Many studies have concluded that terrain attributes are valuable predictors of soil nutrients (Zeraatpisheh et al., 2019, Tajik et al., 2020). Likewise, the relative influence of terrain-derived covariates was distinct for all micronutrients (Fig. 5). Channel networks base level, multi-resolution ridge-top flatness index, and channel networks distance were the most influential terrain parameters. Yet, the relative superiority of climatic variables over terrain parameters can stem from the fact that the elevation was somewhat uniform across these agro-climatic regions and therefore was not the most dominant controlling factor. Furthermore, the spatial variability of relatively immobile micronutrients is governed by their solubility in soil solution. Geochemical drivers like redox potential, pH, soil minerals, organic matter, and nutrient interactions mainly determine micronutrient availability in soil solution (Dhaliwal et al., 2019). Thus, climatic variables should be integral to future geospatial soil micronutrient exploration. Among remote sensing-based variables, EVI followed by NDVI, was influential for Cu, Fe, and Mn content. For Zn, bare soil index and coloration index showed the highest impact. Among other covariates, the clay index appeared influential for Zn and Cu, while hue indexes were important for Fe and Mn. These results were consistent with the previous studies for other nutrients (Abdel-Kader, 2011; Elhag and Bahrawi, 2017; Taghizadeh-Mehrjardi et al., 2019). Notably, although pH followed by SOC appeared influential in explaining Fe variability, no significant linear correlation was observed between pH, SOC, and all four targeted micronutrients (Fig. SM-3).

Notably, ensemble learning models can be useful in cases where a single tree-based model cannot capture the full complexity of the data or there is a significant amount of noise in the data. Ensemble learning can also be helpful in situations with limited data available (Fan et al., 2018), as it can help mitigate the effects of data sparsity and improve the generalization performance of the model. By using multiple models, ensemble learning can also help identify areas of uncertainty in the data, allowing for better-informed decision-making. Additionally, ensemble learning can help identify and mitigate biases in the data by ensuring that any one feature or variable does not overly influence the model.

However, it is important to note that sometimes ensemble learning can be computationally expensive and may require a significant amount of data to train the individual models. Therefore, it may not be practical for all applications or datasets. Additionally, it is important to carefully select the individual models that are included in the ensemble, as poorly

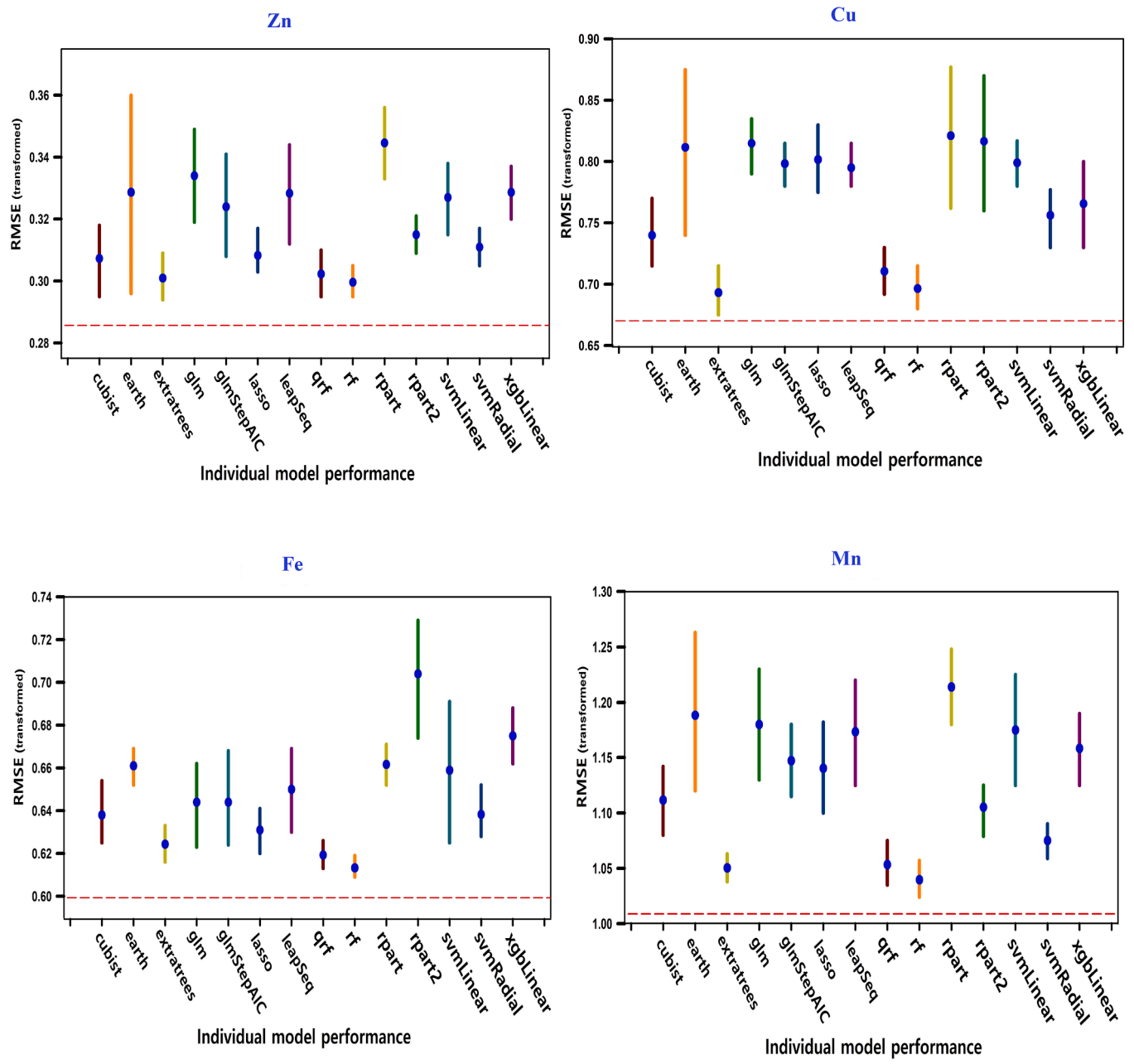


Fig. 3. The comparison of average validation root mean square error (RMSE) values of all 14 base learners. The red dashed line indicates the RMSE value derived from the ensemble model. All the models were executed with transformed values of soil micronutrients using the TukeyKrammer approach to compare their efficiency. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

performing models can actually decrease the overall performance of the ensemble. In summary, while high-resolution maps and ensemble learning are powerful tools for improving the accuracy and robustness of models, their use should be carefully considered based on the specific characteristics of the data and the available computational resources. More research is necessary to determine the required granularity vs. processing power and memory.

3.3. *Partial dependence plots of influential variables*

Partial dependence plots (PDP) of the top four influential covariates (Fig. 6) demonstrated the marginal effect of each independent covariate on the available micronutrient (transformed values) while the effects of other covariates were kept constant (Friedman, 2001). The general trend of the influencing patterns by every covariate varied between micronutrients. Precipitation seasonality, which quantitatively expressed the high-energy rainfall at the end of the summer or beginning of autumn, was strongly influential for micronutrient availability under water-logged rice cultivations. Specifically, PDP indicated a positive relation between precipitation seasonality with available Zn and Cu, with a sharp increase between 90 and 100%, representing a long dry season. This trend, however, slowed down for much higher precipitation seasonality (>100%) at a varying rate. Given that severe Zn deficiency sometimes occurs under low temperatures (Brennan et al., 1993), available Zn content increased sharply with the mean temperature of both the warmest and wettest quarters. A similar pattern was observed with the annual mean temperature for available Mn, especially beyond 27 °C. The strong influence of soil pH on soil biogeochemical processes (Rengel, 2015) was evident from its direct and negative association with available Cu and Fe. For Mn content, a sharp fall was observed in the

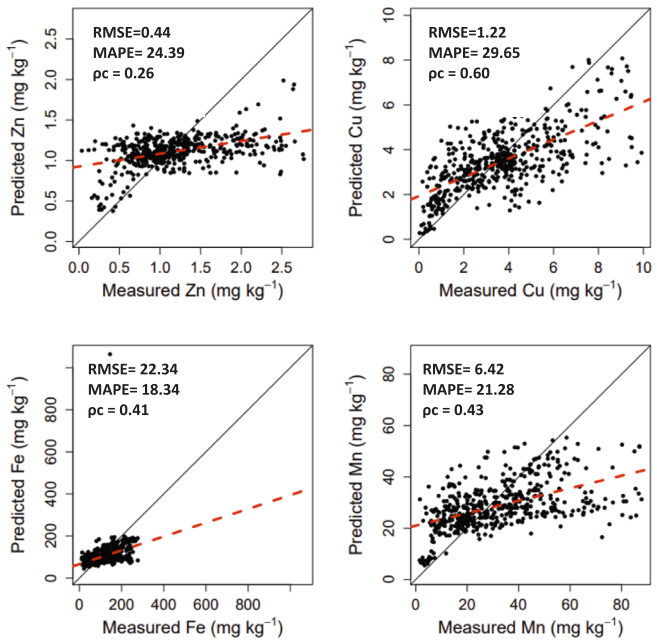


Fig. 4. Lab-measured vs. ensemble model-predicted values of four tested micronutrients of the Indo-Gangetic Plain (withheld 30% data set) using the back-transformed values. The solid line and the red dashed line indicate the 1:1 line and the regression line, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

alkaline range (pH > 7). A similar pattern of steady increment of Fe and Mn resulted in annual precipitation up to 1650 mm, and after that, a steep increase was found. Najafi-Ghiri et al. (2013) also observed high Fe under aquic moisture regimes in calcareous soil. The intense and frequent rainfall pattern of CSZ can explain the elevated level of soil micronutrients. Indeed, precipitation influences several geochemical reactions under a specific pH range which changes metal solubility from soluble (e.g., Zn^{2+}, Fe^{3+}, and Mn^{2+}) to insoluble phases (e.g., Zn-carbonate, Fe, and Mn hydroxides) (Sauvé et al., 2000; Moreno-Jiménez et al., 2019). Among remote sensing-based indices, EVI, related to vegetation greenness, exhibited a positive association with available Mn concentration. Vegetation indices had the potency to explore the soil variability due to the close association between soil properties and vegetation (Falahatkar et al., 2016). The positive effect of SOC on available Fe (Fig. 6) can be attributed to the chelating ability of SOM (Gungor and Let, 2010) and the production of organic acids, which offer favorable pH for micronutrient availability (Zhao et al., 2018).

3.4. Mapping micronutrients content and their uncertainty

Spatial maps of micronutrient concentration of the study area, using back-transformed values, were generated at 150 m spatial resolution (Fig. 7). Available Zn content increased from north and northwest (NAZ) to south and southwest (CSZ, VAZ, and RLZ), ranging from < 0.9 mg kg^{-1} to > 1.3 mg kg^{-1}. Except for a patch in the middle, available Cu content varied between < 2.6 to > 5.8 mg kg^{-1} and from the northern to the southern side of the study area, Cu concentration increased. A similar increasing trend from north to south was observed for available Fe, indicating the higher Fe concentrations in the CSZ, possibly due to microbial sulfate reduction from periodic seawater inundation and subsequent increase in bioavailable Fe (Ling et al., 2015). In the western part (RLZ), lower soil pH, higher mean temperature in both the warmest and wettest quarters, higher bare soil area because of low precipitation, and a relatively high terrain was associated with the elevated level of soil Zn and Fe. In addition, the intense humid tropical climate is more weathered and leached; favoring reddish-brown laterite formation, and featuring Fe or Al oxide-based mineral assemblages (Ghosh and Guchhait, 2015). The NAZ, which is under intensive agricultural use of high-yielding rice varieties exhibited comparatively lower concentrations of Cu (2.6–3.9 mg kg^{-1}) and Fe (80.5–122.9 mg kg^{-1}). This may result from increasing crop yield potential and the high rate of nitrogenous fertilizer applications without organic manuring (Jones et al., 2013).

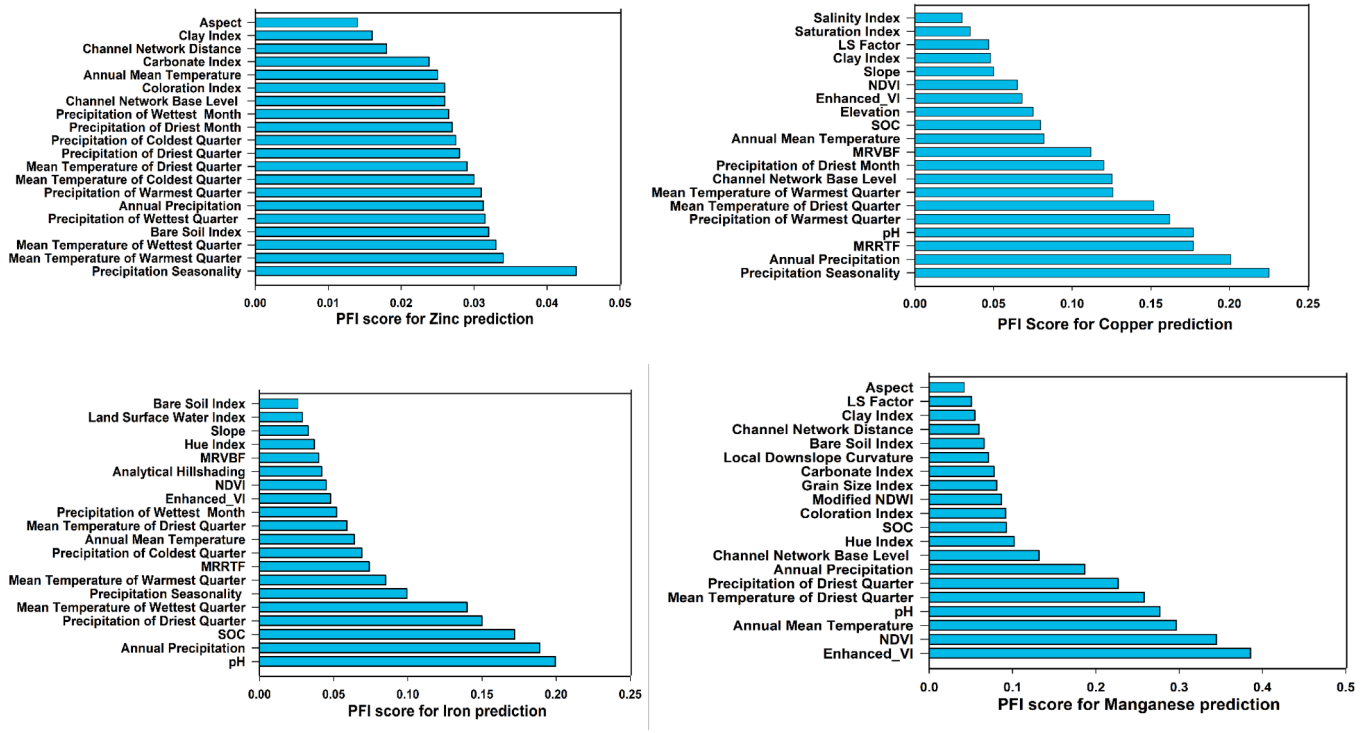


Fig. 5. Plots showing relative variable importance for predicting four micronutrients of the Indo-Gangetic Plain.

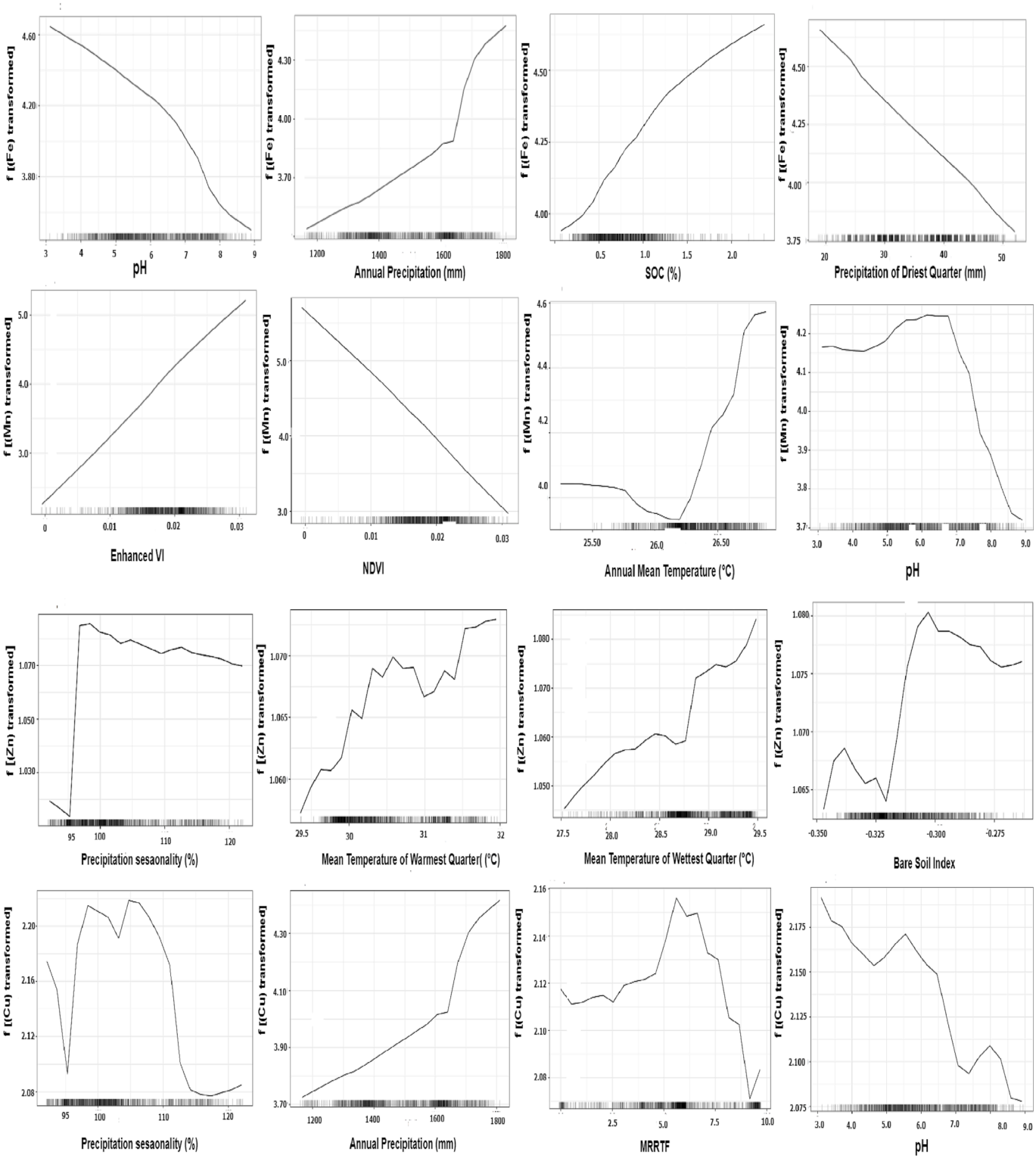


Fig. 6. Partial dependence plots (PDP) of the top four influential covariates for predicting four micronutrients (transformed values) of the Indo-Gangetic Plain. PDP shows the marginal effect of a variable (or two variables) on the fitted model after accounting for the average effects of the other variables in the model. The trend and shape of the function is much more informative than the values themselves on the vertical axis. The tick marks at the base of the plot are deciles of the input variables (e.g. 10%, 20%, …, 90%).

The VAZ exhibited comparatively lower concentrations of available Fe. For available Mn, relatively low concentrations were observed in the NAZ. However, available Mn content was higher at the western patch in the VAZ. Nonetheless, the mean value of available soil micronutrient contents in all four agro-climatic zones was above their regional critical threshold limits (Tandon, 2005; Dasgupta et al., 2022), which is a common occurrence in wetland rice production in Southeast Asia (Sahrawat, 2008).

While mapping the spatial distribution of the 90% prediction limit range, both available Zn and Fe maps exhibited lower prediction

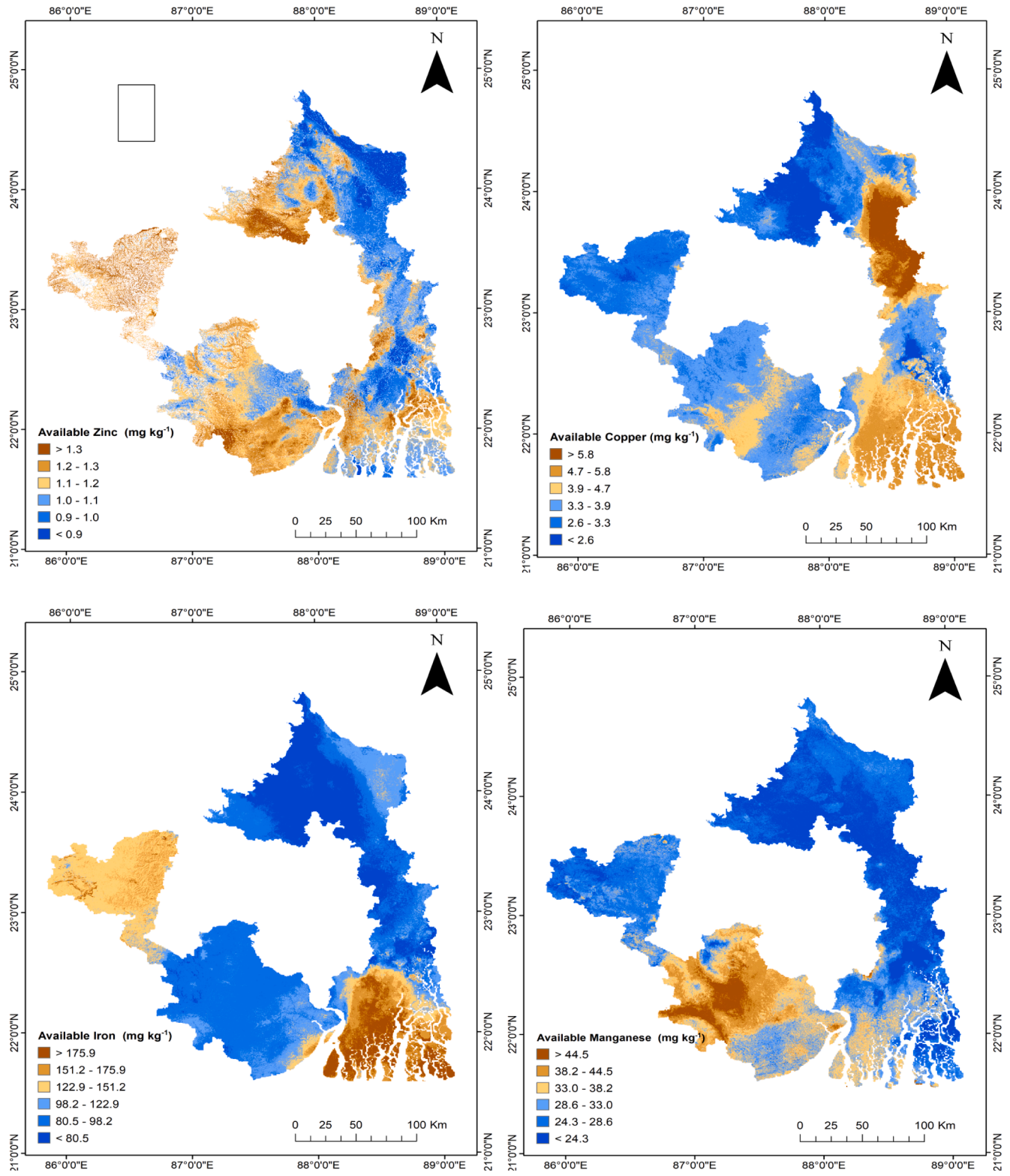


Fig. 7. Spatial maps of micronutrients of the from the study region under the Indo-Gangetic Plain.

uncertainties and narrower prediction limit ranges (0.2–0.5 mg kg^{-1}, and < 44.5 mg kg^{-1} for available Zn and Fe, respectively) (Fig. 8). Conversely, for Cu, an area in the NAZ exhibited a comparatively higher prediction limit range (>8.1 mg kg^{-1}) which can be attributed to the lower sampling density due to the vicinity of this area to the India-Bangladesh international border. Moreover, the prediction limit range for available Mn was comparatively lower in the VAZ, presumably because of lower sampling density. Accordingly, future sampling plans should focus on collecting more samples in these areas to reduce the existing prediction uncertainty.

3.5. Iron and zinc biofortification in rice

Digital micronutrient maps based on ensemble modeling can help

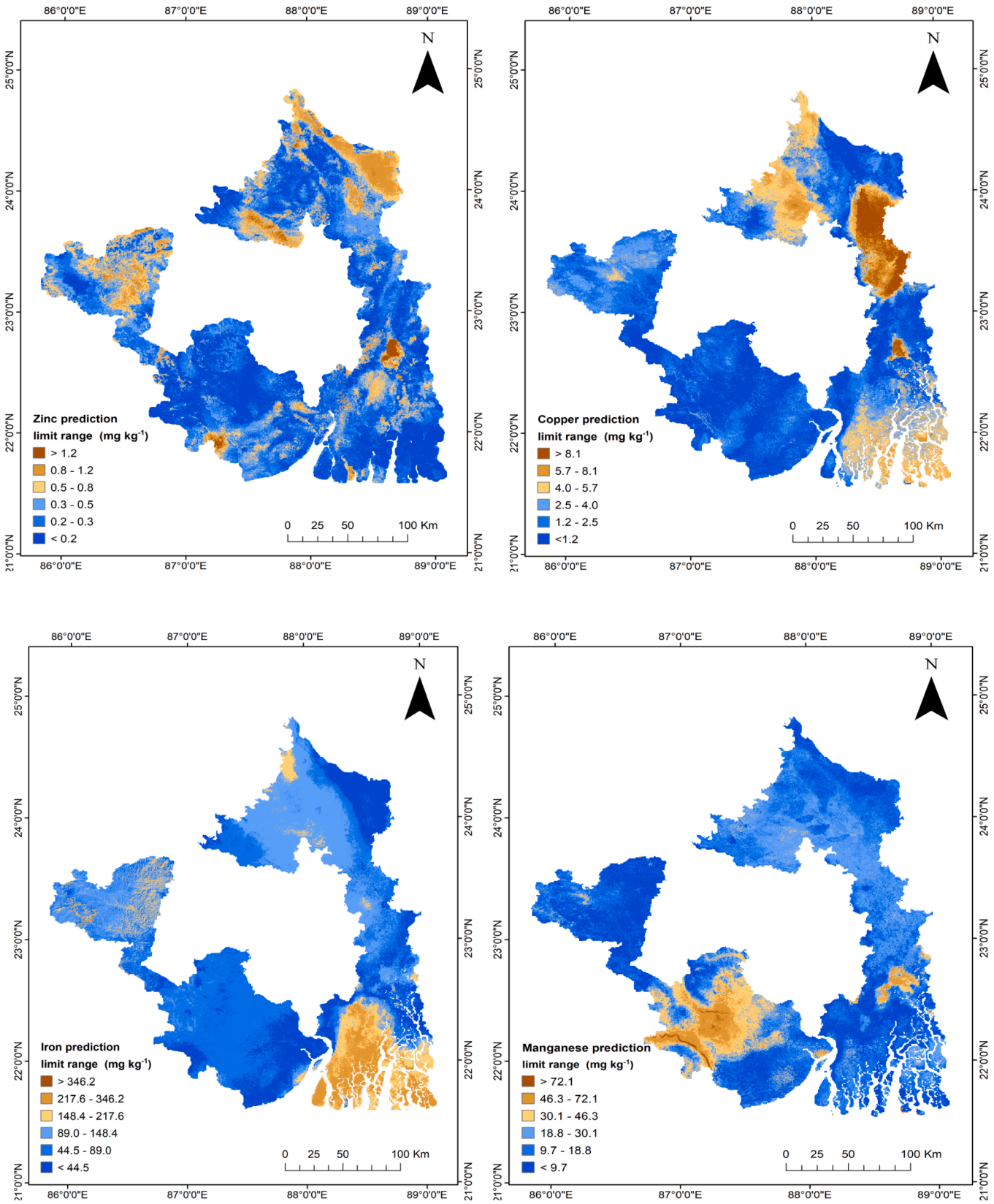


Fig. 8. Spatial distribution of 90% prediction limit range, computed based on bootstrap calibration data, of micronutrients from the study region under the Indo-Gangetic Plain.

identify intervention hotspots and guide management decisions. Notably, available soil Fe in the entire study region was much higher than the deficiency threshold level for rice (6.0 mg kg^{-1}) (Basu, 2011). Likewise, available Zn ranged from medium to adequate levels for rice growing and varied significantly with changes in fundamental soil chemical properties. Yet, due to diverse soil and environmental conditions, high available soil nutrient concentration does not always reflect higher crop uptake. Therefore, exploring the inter-relationship between micronutrient concentration in soil and crop grain is required to understand the biofortification potential for any place. In this study, Zn content in rice grain ranged from 10.22 to 30.02 mg kg^{-1} (mean of 23 mg kg^{-1}) and Fe content varied between 51.18 and 143.44 mg kg^{-1} (mean of 102.65 mg kg^{-1}), respectively. Hence, grain Zn concentration was below the critical threshold value (28–30 mg kg^{-1}) (Saha et al., 2017). Conversely, Zn contents in rice straw ranged from 17.35 to 49.98 mg kg^{-1} with a mean of 35 mg kg^{-1}. Fe contents in rice straw also varied widely from 113.15 to 279.9 mg kg^{-1}, with a mean of 199.44 mg kg^{-1}. Grain concentrations of Fe were found to be higher than the critical limits. The linear regression models revealed moderate agreement between soil available and grain micronutrient concentrations, with R^2 values of 0.52 to 0.63 for Zn and Fe, respectively (Fig. 9). Similar agreement was observed for rice straw, with R^2 values of 0.59 to 0.68 for Zn and Fe, respectively. The developed models were used for spatially predicting grain Zn and Fe content from their respective soil concentration in parts of the VAZ (Fig. 10). Results revealed that most areas in the VAZ had low grain Zn level, that is less than the targeted Zn level in rice grains. However, in the case of grain Fe content, areas of the eastern part of the VAZ, covering South 24 Parganas, exhibited higher levels of grain Fe than other parts. The spatial variation map of grain Zn and Fe can help in identifying specific pockets where rice varieties with biofortification potential can be planted. This will also enable identifying areas where specific mitigation strategies should be taken for enhancing the micronutrient content in rice grains. However, rice grain processing causes a considerable loss of Zn and Fe (>60%) and eventually results in minimal human consumption (Saha et al., 2015).

The relevance of effective biofortification strategies in effectively transferring micronutrients from the field to the human gut has gained worldwide acceptance. The significant increment in grain concentration and yield by micronutrient fertilization in cereals has been previously reported in India (Mandal and Mandal, 1990). In an earlier experiment in the same region, a 53% improvement (0.9 mg d^{-1}) in human Zn consumption was reported when fed with Zn biofortified cooked rice compared with nonfortified rice (Saha et al., 2017). Recently, the Indian government launched the National Nutrition Mission for staple food fortification as a cost-effective approach to control vitamin and mineral deficiencies (Press Information Bureau, 2018). In addition, the Integrated Child Development Services (ICDS) scheme was launched in 1975 for food distribution in government hostels and hospitals. Moreover, India's Public Distribution System (PDS) has enormous potential to reach almost 100% of the malnourished and poor. Joy et al. (2015) demonstrated the capability of agronomic biofortification through Zn-enriched fertilizers to alleviate human dietary Zn deficiency despite low soil Zn availability. While soil and foliar application is recommended as the most effective method (Cakmak et al., 2010), indiscriminate and blanket fertilization must be avoided due to a narrow range of deficiency and toxicity for micronutrients in plants. For example, blanket Zn application in high-Zn soils sometimes reduces grain Zn and yield due to the antagonistic relationship with P and other nutrients (Mandal and Mandal, 1990). Saha et al. (2017) reported that the percentage loss of applied Zn during the processing operations was maximum (28.7 mg kg^{-1}) when used with the highest recommended dose (soil + two foliar applications). Consequently, the nutrient recommendation is based on soil test values. Although several soil test kits have been developed for many soil parameters, such tests are not accurate or sensitive enough to assess regional micronutrient status. Thus, this study has proposed a coordinated regional strategy (Fig. 11) to minimize the cost of extensive experimentation and soil analysis for biofortification planning and judicious fertilization. Succinctly, the DSM-generated micronutrient maps and crop grain analyses can determine the zone-specific input and, therefore, intervention needed. This will help policymakers with large-scale decision-making, facilitating investment opportunities and sustainable micronutrient recommendations toward micronutrient-enriched food. In addition, this holistic platform can provide app-based digital fertilizer recommendations, especially in densely populated developing countries where farm soil management practices vary widely.

It is well known that micronutrient biofortification depends on both soil micronutrient status and the genetic potential of the crop variety (Cakmak and Kutman, 2018). A crop variety with high biofortification potential cannot perform well under deficient soil micronutrient content. Therefore, knowing the soil's inherent potential for properly planning fertilizer application is imperative. Similarly, high soil micronutrient content does not necessarily induce high plant uptake. Consequently, modeling plant micronutrient uptake and its translocation to grain is complex. The present study followed a simple modeling scheme to find a geographical link between inherent soil fertility and grain micronutrient content. Though the relationship is weak, the proposed approach can help identify potential regions where high soil Zn/Fe concentrations are also reflected in their higher content in grains or vice-versa. In the present study, the feasibility of transferring soil micronutrient concentration into grain content in parts of VAZ was evaluated. However, the cost of yearly mapping micronutrient concentration makes it infeasible for yearly modeling of micronutrient content in crop grains. The model should be recalibrated once new crop varieties are introduced in a significant percentage of the area under consideration. The soil-grain micronutrient content relationship may vary in other agroclimatic regions because of changes in crop variety or other environmental factors. Hence, future research should extend the modeling over other agroclimatic regions.

Admittedly, in the progress of data-driven soil mapping, the fact that the soil surface is covered during most of the year can be seen as a limitation, especially in agricultural production areas (Tziolas et al.,

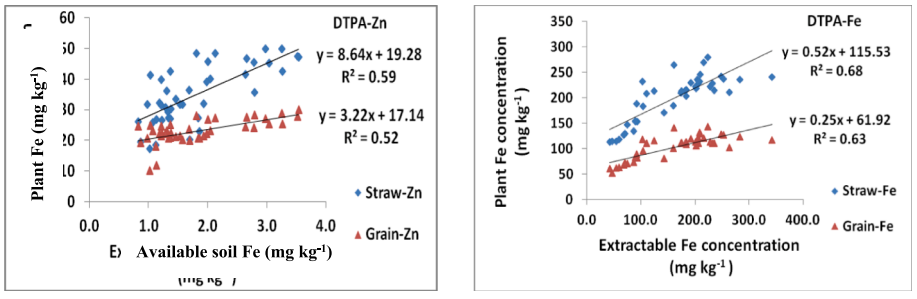


Fig. 9. The linear regression models between soil available and grain/straw micronutrient concentrations for samples from the study region under the Indo-Gangetic Plain.

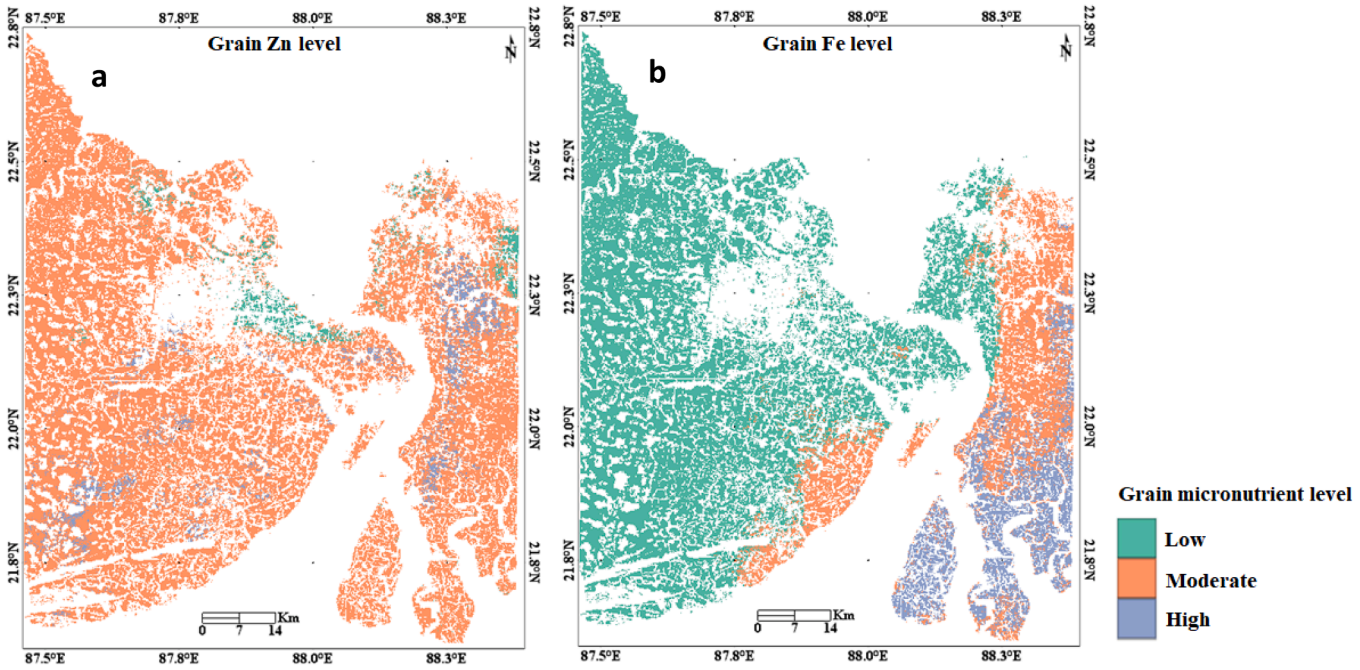


Fig. 10. Spatial variations of (a) grain Zn and (b) grain Fe content over rice crop fields in part of Vindhyachal Alluvial Zone.

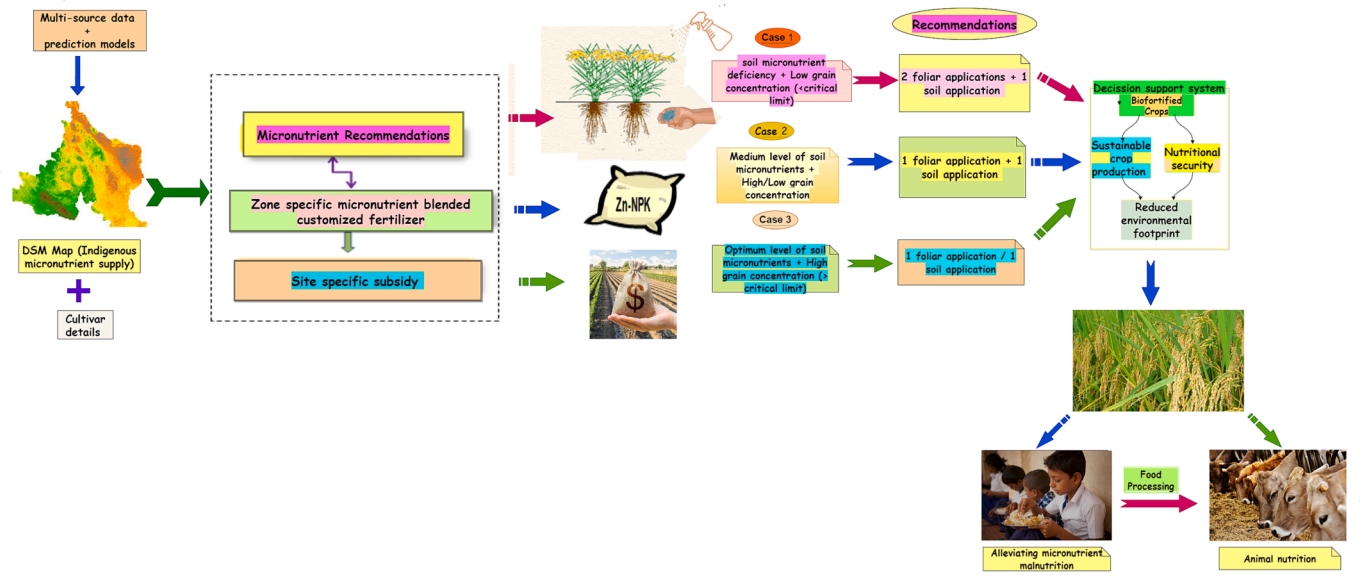


Fig. 11. A coordinated regional strategy for biofortification planning and judicious fertilization using digital soil maps of micronutrients.

2021). Moreover, integrated scenarios combining Landsat 8 and Sentinel-2 remote sensing data, climate variables, and machine learning algorithms should be evaluated for exploratory studies in similar areas without soil sampling.

3.6. Using DSM for popularizing micronutrient enriched custom blends and fortified fertilizers and direct benefit transfer

Several technical and socio-economic policies prioritize more accessible and affordable micronutrient-enriched fertilizers worldwide. At competitive prices, regional or zonal soil-crop-specific customized blending and micronutrient fortification can supplement traditional soil test-based recommendations. For example, in 2008, the Indian government issued customized fertilizer policy guidelines for manufacturing and sale, and in 2010, a product pricing system was formed for a nutrient-based subsidy (NBS) regime. The NBS scheme targeted new product innovation and the production of more grades of blended micronutrient fertilizers at subsidized rates. Nevertheless, after ten years of implementation, the total production and use remain low (Kishore et al., 2021). Hence, popularizing and manufacturing these custom blends with recommended nutrient doses need more reliable, robust, high-density sampling-based soil test information.

This study shows that multi-source data fusion with an ensemble model and digital micronutrient mapping approach can be a reasonable option to depict a zone-specific quantitative representation of soil micronutrients. Such information can guide agri-business and policy-makers to target their interventions. Manufacturers can use these DSM maps to prepare regional or zonal fertilizer grades and improve their

existing and potential marketing in target areas. In addition, developing countries can use the DSM micronutrient map to fix the subsidy amount to manufacturers for micronutrient-fortified fertilizers, as per NBS. Economically, soil mapping initiatives and improved soil test-based fertilizer recommendations are highly remunerative in India, with a potential benefit-to-cost ratio of 17:1 (Wani et al., 2016). These would offer a digital soil intelligence platform for countries like India, with a long-term strategy to shift subsidies from manufacturers to farmers to reduce irregularities under the direct benefit transfer scheme (Kishore et al., 2021).

4. Conclusions

This study used a hybrid ensemble ML-based DSM technique for spatial prediction and mapping of four available soil micronutrients. Subsequently, this study has proposed a coordinated regional strategy to minimize the cost of extensive experimentation and soil analysis for biofortification planning and judicious fertilization. The study has been comprehensively summarized as follows:

The ensemble model reasonably predicted micronutrient concentrations in the IGP better than 14 base learners.

Regional-scale micronutrient maps can reduce gaps in existing DSM literature by producing better maps of available soil micronutrients, identifying intervention hotspots, and guiding management decisions.

Micronutrient maps can identify areas where crop varieties with biofortification potential can be planted.

The results of this study can improve the understanding of influential factors for spatial variability of soil micronutrients and serve as regional-scale micronutrient base maps to examine biofortification potential and devise economical agri-business policies.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.geoderma.2023.116457.

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